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## SECOND REVIEW UNDER THE POLICY SUPPORT INSTRUMENT—DEBT SUSTAINABILITY ANALYSIS

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*This debt sustainability analysis (DSA) updates the joint IMF/IDA DSA from June 2013.*[2] *Mozambique remains at moderate risk of external public debt distress: since the last DSA there has been significant new non-concessional borrowing and the discount rate has been updated from 3 percent to 5 percent, but key indicators continue to remain below the thresholds in the baseline scenario. While staffs agree with the authorities on the importance of public investment for development and that it will need to be financed by at least some non-concessional borrowing, the potential increased risks to debt sustainability should be contained by moderating public external borrowing from its recent pace and through further improvements in debt management and investment planning capacity. As public debt is largely external, the evolution of total public debt indicators mirrors that of public external debt. Private external debt is expected to increase rapidly in importance, driven by investment in the natural gas sector, and to comprise the majority of external debt by the end of this decade. The authorities were in broad agreement with the DSA outlook and presentation. Their technical comments have been taken on board.*

[2]The DSA presented in this document is based on the standard low-income countries (LIC) DSA framework. See "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (http://www.imf.org/external/pp/longres.aspx?id=4827) and World Bank Report No. ACS6956, 10/23/13). Under the Country Policy and Institutional Assessment (CPIA), Mozambique is rated as a medium performer, albeit close to the threshold of 3.75 for strong performers, with an average rating of 3.71 during 2010–12; the DSA uses the indicative threshold for medium performers.

# UNDERLYING DSA ASSUMPTIONS

**1. This DSA is consistent with the macroeconomic framework outlined in the Staff Report for the Second Review under the PSI** (Box 1). Compared to the previous DSA,[3] the main changes in this DSA are as follows:

**a. The medium-term macroeconomic framework has been updated but is similar to the last DSA** (Text Table 1). Economic growth was dampened in 2013 as a result of flooding early in the year that damaged agricultural output and transport infrastructure. Growth is projected to bounce back quickly and accelerate over the medium term as agriculture recovers, extractive industries boom, and infrastructure investments materialize. The fiscal deficit was lower in 2013, among other factors because of windfall capital gain tax revenues received. The current account deficit is projected to remain in the range of 39-41 percent of GDP over the medium term reflecting significant imports of goods and services related to the development of the gas and coal sectors. These imports are financed primarily by non-debt creating FDI.

**Text Table 1: Evolution of Selected Macroeconomic Indicators Between DSA Updates**

| | 2012 | 2013 | 2014 | 2015 | 2016 |
|---|---|---|---|---|---|
| | | | Projections | | |
| Real GDP growth (%) | | | | | |
| Previous DSA | 7.4 | 7.0 | 8.5 | 8.5 | 8.0 |
| Current DSA | 7.2 | 7.1 | 8.3 | 8.2 | 8.2 |
| Nominal GDP (US$ billion) | | | | | |
| Previous DSA | 14.2 | 14.7 | 16.0 | 17.9 | 20.1 |
| Current DSA | 14.3 | 15.3 | 17.1 | 18.7 | 20.8 |
| Overall fiscal deficit (% of GDP) | | | | | |
| Previous DSA | 4.2 | 6.7 | 7.2 | 6.7 | 6.7 |
| Current DSA | 4.1 | 2.8 | 9.2 | 7.5 | 6.6 |
| Current account deficit (% of GDP) | | | | | |
| Previous DSA | 36.5 | 39.9 | 41.3 | 41.4 | 39.8 |
| Current DSA | 45.4 | 39.5 | 46.9 | 47.0 | 43.8 |
| FDI (% of GDP) | | | | | |
| Previous DSA | 36.6 | 32.2 | 25.8 | 20.8 | 18.9 |
| Current DSA | 20.7 | 36.5 | 33.0 | 28.6 | 26.4 |
| Use of NCB ceiling (% of GDP) | | | | | |
| Previous DSA | 0.8 | 2.4 | 2.5 | 1.6 | 1.3 |
| Current DSA | 0.8 | 7.8 | 2.4 | 1.9 | 1.7 |
| Disbursements under PTL (% of GDP) 1/ | | | | | |
| Previous DSA | 0.4 | 0.2 | 0.3 | 0.2 | 0.2 |
| Current DSA | 0.4 | 0.2 | 0.3 | 0.2 | 0.2 |

1/ Portuguese credit line.

[3] See IMF Country Report No. 13/200.

### Box 1. Macroeconomic Assumptions 2014–34

*The medium-term assumptions in the baseline scenario for 2014-34 are consistent with the medium-term macroeconomic framework underlying the Staff Report for the Second Review under the Policy Support Instrument.*

**Real GDP growth** is projected to be around 8 percent on average in the longer term. Growth is expected to accelerate over the medium term, supported by recovery in agricultural production, expansion in coal mining, and infrastructure investments, including support for coal exports and LNG manufacturing. A sharp increase in growth in 2020 reflects the assumed coming on line of the first natural gas "train" and thus exports in that year. Growth is sustained in the long term by strong population/labor force growth, continued infrastructure investment, and related productivity gains. Risks to growth include public and private investment not achieving expected payoffs and thus limiting productivity gains, and the possibility of Dutch disease effects.

**LNG sector.** LNG plants are assumed to be under construction during 2014–22. The projection assumes a moderate-sized plant consisting of four LNG manufacturing units ("trains"). One train is assumed to start production in 2020, followed by a second train in 2021, and the third and the fourth train will start production in 2023. Total investment is projected at $40 billion. The sector's contribution to GDP is expected to be small during the construction period due to high import content. Annual LNG output will reach 20 million tons in 2023, contributing more than 20 percent of nominal GDP by then.

**Consumer price inflation** is projected to remain in the authorities' target range of 5–6 percent over the medium term.

**Growth of exports** is projected to stabilize at around 4 percent a year in the longer term as coal and LNG exports stabilize. In the short term export growth rates show sharp changes as a result of large coal and LNG development projects. In particular the growth rate of exports would almost double in 2020-23 due to LNG exports coming on line.

**Imports** are projected to show abrupt jumps in 2014-2017 during the LNG plant construction phase but their growth would stabilize at around 7-8 percent a year in the long term.

**The non-interest external current account deficit** is projected to rise to over 40 percent of GDP in the medium term largely driven by LNG-related imports. The deficit will be primarily financed through FDI and private external borrowing. It would then average 14 percent of GDP in the long term as coal exports increase with transport capacity and as LNG exports start.

**The non-interest primary fiscal deficit** is projected to widen in 2014, reflecting an expansionary budget in an election year, including one-off expenditure on maritime security (2.1 percent of GDP). The fiscal balance would improve in the medium to long term as revenue efforts continue, the wage bill and public investment taper off to more sustainable levels and other non-interest current expenditure falls in percent of a fast-growing GDP. The fiscal balance is expected to improve further beyond 2020 once LNG revenue commences.

**b. A revised public investment and external borrowing profile.** Non-concessional borrowing was significantly higher in 2013 than previously projected because of the government guarantee for Loan Participation Notes (LNP) of $850 million (6 percent of GDP) issued by the newly created enterprise EMATUM (tuna fishing). With this, total non-concessional borrowing in 2013 amounted to almost 8 percent of GDP. Other projected disbursements of concessional and non-concessional debt are similar to the previous DSA (Text Table 2).

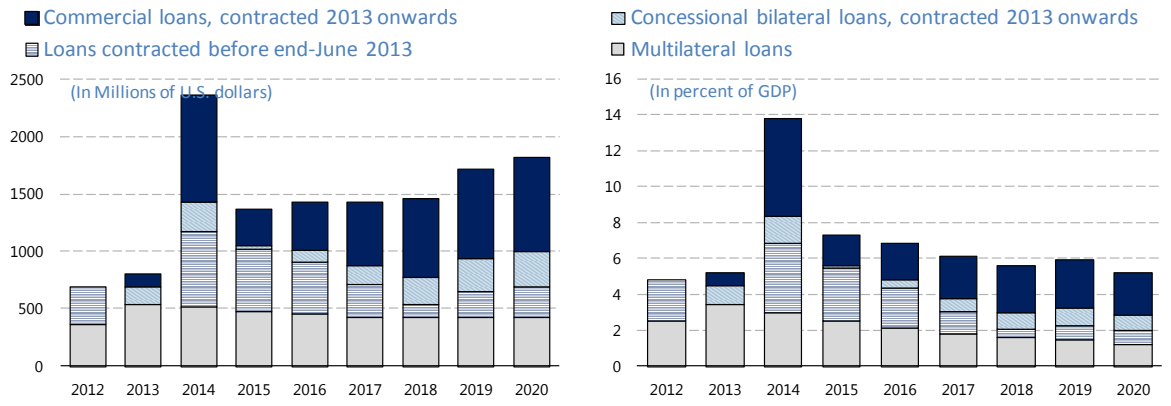
**Text Table 2. Non-concessional Loans Contracted and in the Pipeline 2013-2014**

| | Signed on | Loan amount (US$ millions) | Before 2nd rev. | Before end-2014 |
|---|---|---|---|---|
| **Total non-concessional loans** | | **1,454** | **926** | **528** |
| Brazil | | | | |
| Nacala Airport (Augmentation) | July, 2013 | 45 | 45 | |
| Public Transport Maputo-Matola | Expect to sign by end-2014 | 135 | | 135 |
| Nacala Industrial Free Trade Zone | Expect to sign by end-2014 | 40 | | 40 |
| Dam Moamba Major | Expect to sign by end-2014 | 353 | | 353 |
| India | | | | |
| Beira Port-Terminals, Storage Tanks, Etc | December 2013 | 31 | 31 | |
| EMATUM* | September 2013 | 850 | 850 | |

* Government Guarantee

**c. A revised grant element in financing terms in the medium term.** The last DSA assumed that the grant element for new borrowing would experience a pronounced decline during the next two decades. The overall pattern is similar in the current DSA (given the decline in the long-run availability of traditional donor financing the assumption is that the government will rely increasingly on commercial borrowing) but grant financing in terms of GDP is somewhat higher in the near term reflecting updated information from donors.

**Text Figure 1. Public Sector External Debt Disbursement Path**



d. **Private sector debt stocks and medium-term borrowing have been revised upwards**. Private external debt is estimated at 24 percent of GDP in 2013, and projected to rise, financing the building of gas liquefaction capacity, to some 90 percent in 2019, but fall back to 44 percent by 2024 when LNG is exported and private borrowing amortized. The accumulation is significant and could pose a risk if the associated investments do not achieve the expected payoffs.[4]

e. **The increase in the DSA discount rate from 3 percent a year ago to 5 percent in this DSA** yields a lower present value of public debt/borrowing in this DSA compared to the previous one, but with the revisions in private debt/borrowing the overall debt is now slightly higher over the medium term than in the last DSA.

# EXTERNAL DEBT SUSTAINABILITY ANALYSIS

**2. Total external debt is projected to rise rapidly during this decade, reflecting mainly private sector investment in the natural gas sector.** External debt (both public and private) is expected to peak at about 142 percent of GDP in 2019, at which time private sector debt will constitute about two thirds of total external debt. With investment in the coal sector projected to be financed through foreign direct investment, this increase in private external debt is mostly driven by investments in the natural gas sector. Public sector debt is expected to peak in 2019 at 33 percent of GDP (in PV terms). The significant build-up of private sector external debt needs to be monitored by the authorities to contain vulnerabilities. With renowned global companies leading investments in the natural resource sector, however, the risk for government contingent liabilities or other vulnerabilities beyond those specific to the natural resource operations is currently considered to be low.[5]

**3. All public external debt indicators remain below their respective thresholds in the baseline** (Figure 1). With the unexpected (at the time of the last DSA) government guarantee of the EMATUM LNP, nominal debt is significantly higher in 2013 than previously projected, but the PV of debt is lower than in the last DSA as a result of the increase in the discount rate. The PV of debt in terms of GDP now peaks around 33 percent (in 2019), compared to a peak of 39 percent in the previous DSA. The beginning of LNG production and the ensuing surge in GDP would drop the ratio down to below 30 percent in the next decade (Table 1).

**4. External debt ratios remain vulnerable to FDI, growth, and exchange rate shocks.**[6] The threshold for the PV of debt to GDP ratio is breached under the most extreme stress tests for a

[4] Further analysis will be conducted at the time of the next review to evaluate the nature and extent of vulnerabilities

[5] Likewise, the risk of BOP pressures emerging as a direct result of megaproject-related investment activity is considered to be low. Care will have to be taken in the long term, however, once revenues from these ventures materialize, as these may be volatile reflecting world commodity prices and in relation to imports. Moreover, large natural resource exports then also hold competitiveness risks emerging from a possible exchange rate appreciation, which will have to be carefully managed.

[6] The impact the of the standard shocks in the DSA template is heightened by the fact that the standard stress tests revert to historical values, which are significantly different from current and expected values because of the structural change in the Mozambican economy resulting from the large-scale exploitation of coal and natural gas since 2011.

sustained period.[7] A sharp reduction in non-debt creating FDI and growth in 2014-2015 would push the PV of debt/GDP ratio well above the 40 percent threshold during the next 15 years, with a peak of about 70 percent (Figure 1). Apart from the FDI/growth shock, a sharp depreciation of the metical in 2014 or historical export growth also lead to breaches of this threshold (Table 2). All these shocks lead to a significant overshooting of the threshold for a sustained period.[8] The impact of these shocks is somewhat less than in the previous DSA primarily because of the increase in the discount rate. In three shock scenarios the PV of debt will also breach the debt/exports threshold of 150 percent of exports.[9]

**5. Ensuring that LNG production materilizes is crucial for Mozambique's debt sustainability.** The LNG sector will contribute significantly to GDP, exports and government revenue. A gradual public investment scaling-up, as under the baseline scenario, anticipating some LNG revenue would be appropriate given Mozambique's infrastructure investment needs. If, however, there are delays in project take-off or LNG production or revenue is much lower than expected, the debt ratios would be higher over the medium to long term.

# PUBLIC SECTOR DEBT SUSTAINABILITY

**6. The evolution of public debt indicators (including domestic debt) mirrors that of the external indicators because of the predominance of external debt** (Table 3 and Figure 2)**.** The medium-term increase in public debt reflects the temporary surge in public investment financed in part by external borrowing on non-concessional terms. The DSA assumes that about 10 years after the start of LNG exports the public sector will dedicate an increasing share of resources to pay back debt. Under the baseline scenario, the PV of public debt remains below 50 percent of GDP throughout, therefore remaining well contained and below the indicative benchmarks that research has linked to increased probability of debt distress.[10]

**7. The public DSA illustrates that it is essential for debt sustainability to moderate the pace of new borrowing as is planned by the authorities.** In other words, the fiscal primary balance will need to be reduced in the medium and long run. For illustrative purposes, the "Fix Primary Balance" scenario[11] shows the impact if the primary balance were fixed at values of its peak years 2013-15, i.e. if such a slowing in new borrowing would not take place,[12] debt ratios would

[7] The scenario in which variables are at their historical levels has been omitted given that it generates negative debt as a result of the large changes in variables in the baseline arising from LNG activities.

[8] The charts in Figure 1 display the stress test with the most adverse outcome in 2023.

[9] This breach, however, should not be overemphasized as a fall in FDI would likely be linked to lower imports rather than higher borrowing as implicitly assumed in the standard shock scenario.

[10] At Mozambique's CPIA rating, the indicative public debt benchmark signaling higher risk of debt distress lies between 56 and 74 percent for the PV of debt-to-GDP ratio. See IMF, 2012, "Revisiting the Debt Sustainability Framework for Low-Income Countries."

[11] See Figure 2 and Table 4 (lines A2).

[12] To provide this illustration, the "Fix Primary Balance" scenario was adapted for this analysis from the standard of fixing the primary balance at 2012, instead fixing the primary balance at its 2013-15 average value in percent of GDP.

(continued)

continue to rise throughout the projection period, quickly elevating risks to debt sustainability, no matter which assumptions are made on LNG production.

# CONCLUSIONS

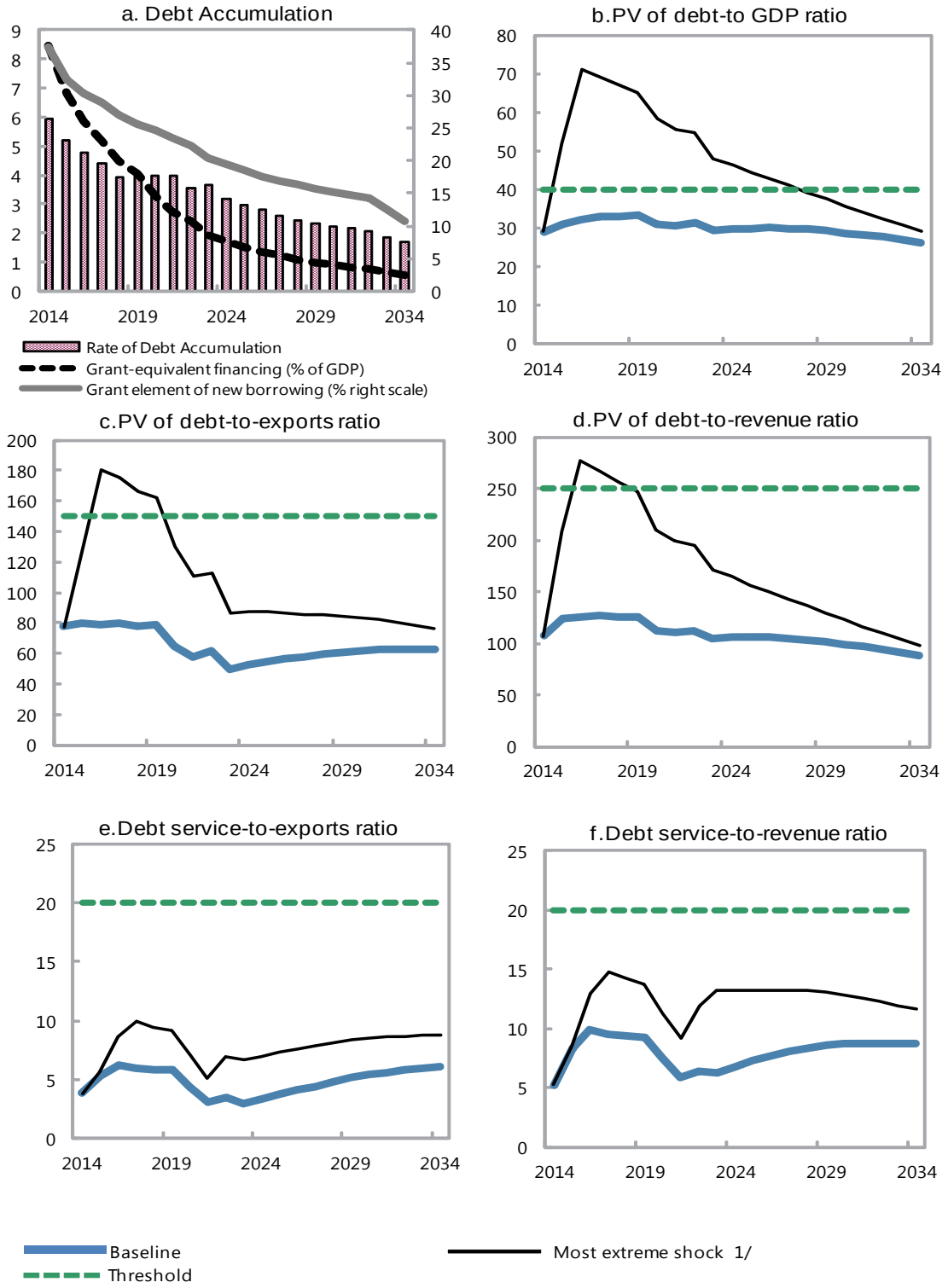
**8. In staffs view, while the standard DSA shocks indicate that Mozambique would be in the moderate risk of external public debt distress category, public investment should continue to move forward but with a more moderate pace of public borrowing.** Even against the background of a temporarily accelerated borrowing pace, the baseline debt trajectories remain below their respective thresholds throughout. Importantly, debt service indicators remain substantially below their thresholds, including under stress tests, and reflect conservative assumptions regarding the grant element for future borrowing.[13] The breaches under the stress tests would be reversed by the coming on stream of LNG production, and seem manageable against the backdrop of the authorities' strong track record of prudent economic management. Moreover, the impact the of shocks is heightened by the fact that the standard stress tests revert to historical values, which are significantly different from current and expected values because of the structural change in the Mozambican economy resulting from the large-scale exploitation of coal and natural gas.

**9. This analysis highlights three important points for debt sustainability.** First, it will be important to continue to improve debt management and investment planning capacity to ensure that the most deserving public investment projects are selected and yield their desired payoff. Second, it will be important for the authorities to moderate the pace of new borrowing to maintain debt sustainability, in line with their commitments. Third, it is becoming more important—including from a debt sustainability perspective—to ensure that LNG production materializes in order to lock in the beneficial effects on GDP and fiscal revenue.

This change was implemented, because the 2012 primary balance is low compared to those in the near future for two reasons: (i) because of windfall revenues in 2012, and (ii) because major investment projects are only starting.

[13] Moreover, these indicators do not account for other buffers, such as comfortable levels of international reserves relative to non-megaproject imports.

## Figure 1. Mozambique: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014-2034 1/




Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Non-debt flows shock; in c. to a Combination shock; in d. to a Non-debt flows shock; in e. to a Combination shock and in figure f. to a Combination shock

**Table 1 .Mozambique: External Debt Sustainability Framework, Baseline Scenario, 2011-2034 1/**

(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical 6/ Average | Standard 6/ Deviation | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | | | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2014-2019 Average | 2024 | 2034 | 2020-2034 Average |
| **External debt (nominal) 1/** | **57.3** | **57.5** | **68.2** | | | **81.4** | **95.9** | **105.9** | **123.6** | **135.8** | **142.3** | | **85.6** | **47.7** | |
| *of which: public and publicly guaranteed (PPG)* | 32.9 | 36.9 | 44.2 | | | 49.2 | 51.4 | 52.3 | 52.4 | 51.6 | 51.0 | | 41.4 | 32.8 | |
| Change in external debt | -21.1 | 0.2 | 10.6 | | | 13.2 | 14.5 | 10.0 | 17.7 | 12.2 | 6.4 | | -6.1 | -2.5 | |
| Identified net debt-creating flows | -15.1 | 1.8 | 2.6 | | | 13.2 | 14.5 | 12.6 | 21.1 | 17.3 | 13.5 | | -2.9 | -8.2 | |
| **Non-interest current account deficit** | **22.9** | **44.0** | **38.0** | **17.4** | **13.4** | **44.8** | **44.5** | **41.2** | **49.4** | **49.4** | **43.7** | | **15.5** | **5.9** | 14.8 |
| Deficit in balance of goods and services | 29.7 | 51.2 | 48.0 | | | 52.3 | 51.2 | 45.3 | 50.7 | 49.3 | 42.3 | | 3.6 | 6.4 | |
| Exports | 30.8 | 33.3 | 36.0 | | | 37.6 | 39.1 | 41.1 | 41.2 | 42.2 | 42.0 | | 57.0 | 42.1 | |
| Imports | 60.5 | 84.5 | 84.1 | | | 89.9 | 90.4 | 86.3 | 91.9 | 91.5 | 84.3 | | 60.6 | 48.5 | |
| Net current transfers (negative = inflow) | -6.9 | -5.8 | -8.5 | -7.0 | 1.0 | -6.5 | -5.8 | -5.3 | -5.1 | -4.9 | -4.7 | | -3.6 | -3.2 | -3.5 |
| *of which: official* | -6.3 | -5.0 | -7.2 | | | -7.5 | -6.8 | -6.4 | -6.1 | -5.9 | -5.7 | | -4.7 | -4.2 | |
| Other current account flows (negative = net inflow) | 0.0 | -1.4 | -1.5 | | | -1.1 | -0.9 | 1.2 | 3.8 | 5.0 | 6.1 | | 15.5 | 2.6 | |
| **Net FDI (negative = inflow)** | **-20.7** | **-36.5** | **-33.0** | **-13.2** | **12.7** | **-28.6** | **-26.4** | **-24.1** | **-23.5** | **-25.9** | **-23.3** | | **-14.3** | **-11.9** | -14.5 |
| **Endogenous debt dynamics 2/** | **-17.3** | **-5.7** | **-2.4** | | | **-3.0** | **-3.6** | **-4.5** | **-4.9** | **-6.2** | **-6.9** | | **-4.0** | **-2.1** | |
| Contribution from nominal interest rate | 1.5 | 1.3 | 1.4 | | | 2.1 | 2.5 | 2.6 | 2.6 | 2.6 | 2.5 | | 1.5 | 1.1 | |
| Contribution from real GDP growth | -4.4 | -3.6 | -3.8 | | | -5.1 | -6.1 | -7.1 | -7.5 | -8.8 | -9.3 | | -5.6 | -3.2 | |
| Contribution from price and exchange rate changes | -14.4 | -3.4 | 0.0 | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| **Residual (3-4) 3/** | **-6.0** | **-1.6** | **8.0** | | | **0.1** | **0.0** | **-2.6** | **-3.4** | **-5.1** | **-7.1** | | **-3.2** | **5.6** | |
| *of which: exceptional financing* | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| PV of external debt 4/ | ... | ... | 49.5 | | | 61.3 | 75.6 | 85.8 | 104.1 | 117.2 | 124.4 | | 74.0 | 41.1 | |
| In percent of exports | ... | ... | 137.4 | | | 163.1 | 193.2 | 209.0 | 252.7 | 277.9 | 296.2 | | 129.8 | 97.8 | |
| **PV of PPG external debt** | **...** | **...** | **25.5** | | | **29.1** | **31.1** | **32.2** | **32.8** | **32.9** | **33.2** | | **29.8** | **26.3** | |
| **In percent of exports** | **...** | **...** | **70.8** | | | **77.5** | **79.4** | **78.4** | **79.7** | **78.1** | **79.1** | | **52.3** | **62.5** | |
| **In percent of government revenues** | **...** | **...** | **92.9** | | | **106.7** | **124.3** | **125.3** | **126.5** | **126.0** | **125.9** | | **105.7** | **88.4** | |
| **Debt service-to-exports ratio (in percent)** | **19.4** | **12.0** | **12.0** | | | **14.3** | **14.9** | **15.2** | **12.9** | **12.1** | **11.7** | | **7.7** | **7.2** | |
| **PPG debt service-to-exports ratio (in percent)** | **2.0** | **2.1** | **2.8** | | | **3.8** | **5.3** | **6.2** | **6.0** | **5.8** | **5.8** | | **3.3** | **6.1** | |
| **PPG debt service-to-revenue ratio (in percent)** | **3.0** | **3.1** | **3.7** | | | **5.2** | **8.4** | **10.0** | **9.5** | **9.4** | **9.2** | | **6.8** | **8.7** | |
| Total gross financing need (Billions of U.S. dollars) | 1.0 | 1.6 | 1.4 | | | 3.7 | 4.5 | 4.9 | 7.3 | 7.4 | 7.3 | | 3.1 | -4.2 | |
| Non-interest current account deficit that stabilizes debt ratio | 44.0 | 43.8 | 27.4 | | | 31.5 | 30.0 | 31.2 | 31.8 | 37.2 | 37.3 | | 21.6 | 8.4 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 7.3 | 7.2 | 7.1 | 7.4 | 0.7 | 8.3 | 8.2 | 8.2 | 7.9 | 8.0 | 7.7 | 8.0 | 6.7 | 6.9 | 8.4 |
| GDP deflator in US dollar terms (change in percent) | 22.4 | 6.2 | 0.1 | 5.3 | 9.8 | 2.8 | 1.2 | 2.8 | 3.4 | 3.4 | 3.5 | 2.9 | 2.4 | 2.4 | 2.5 |
| Effective interest rate (percent) 5/ | 2.5 | 2.7 | 2.7 | 2.6 | 0.6 | 3.5 | 3.4 | 3.1 | 2.7 | 2.4 | 2.0 | 2.8 | 1.8 | 2.3 | 2.1 |
| Growth of exports of G&S (US dollar terms, in percent) | 29.7 | 23.1 | 16.1 | 16.0 | 14.4 | 16.2 | 13.9 | 16.7 | 11.9 | 14.3 | 11.0 | 14.0 | 4.9 | 6.9 | 11.6 |
| Growth of imports of G&S (US dollar terms, in percent) | 62.8 | 59.0 | 6.7 | 21.0 | 22.7 | 19.2 | 10.0 | 6.3 | 18.7 | 11.2 | 2.6 | 11.3 | 6.0 | 6.6 | 7.1 |
| Grant element of new public sector borrowing (in percent) | ... | ... | ... | ... | ... | 37.3 | 32.4 | 30.2 | 29.0 | 26.9 | 25.6 | 30.2 | 19.4 | 10.8 | 17.5 |
| Government revenues (excluding grants, in percent of GDP) | 20.8 | 23.3 | 27.5 | | | 27.3 | 25.0 | 25.7 | 25.9 | 26.1 | 26.4 | | 28.2 | 29.7 | 28.7 |
| Aid flows (in Billions of US dollars) 7/ | 1.6 | 1.3 | 1.9 | | | 1.9 | 1.8 | 1.7 | 1.7 | 1.7 | 1.8 | | 1.8 | 2.0 | |
| *of which: Grants* | 1.0 | 0.8 | 0.8 | | | 0.9 | 0.8 | 0.7 | 0.7 | 0.7 | 0.7 | | 0.5 | 0.2 | |
| *of which: Concessional loans* | 0.6 | 0.5 | 1.0 | | | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.1 | | 1.3 | 1.7 | |
| Grant-equivalent financing (in percent of GDP) 8/ | ... | ... | ... | | | 8.4 | 6.8 | 5.9 | 5.2 | 4.5 | 4.1 | | 1.7 | 0.5 | 1.5 |
| Grant-equivalent financing (in percent of external financing) 8/ | ... | ... | ... | | | 60.4 | 55.2 | 51.7 | 50.9 | 49.8 | 47.2 | | 33.2 | 15.4 | 28.4 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Nominal GDP (Billions of US dollars) | 12.5 | 14.3 | 15.3 | | | 17.1 | 18.7 | 20.8 | 23.2 | 25.9 | 28.9 | | 56.8 | 139.7 | |
| Nominal dollar GDP growth | 31.4 | 13.9 | 7.2 | | | 11.4 | 9.5 | 11.2 | 11.5 | 11.7 | 11.4 | 11.1 | 9.2 | 9.5 | 11.2 |
| PV of PPG external debt (in Billions of US dollars) | | | 3.9 | | | 4.8 | 5.7 | 6.6 | 7.5 | 8.4 | 9.5 | | 16.7 | 36.2 | |
| (PVt-PVt-1)/GDPt-1 (in percent) | | | | | | **6.0** | **5.2** | 4.8 | 4.4 | 3.9 | 4.0 | 4.7 | 3.2 | 1.7 | 2.8 |
| Gross workers' remittances (Billions of US dollars) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.1 | 0.2 | |
| PV of PPG external debt (in percent of GDP + remittances) | ... | ... | 25.5 | | | 29.1 | 31.0 | 32.2 | 32.8 | 32.9 | 33.2 | | 29.8 | 26.3 | |
| PV of PPG external debt (in percent of exports + remittances) | ... | ... | 70.6 | | | 77.2 | 79.1 | 78.2 | 79.5 | 77.9 | 78.8 | | 52.2 | 62.3 | |
| Debt service of PPG external debt (in percent of exports + remittance | ... | ... | 2.8 | | | 3.8 | 5.3 | 6.2 | 6.0 | 5.8 | 5.8 | | 3.3 | 6.1 | |

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

**Table 2. Mozambique: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034 (continued)**

(In percent)

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | **2024** | 2034 |
| **PV of debt-to GDP ratio** | | | | | | | | |
| **Baseline** | 29 | 31 | 32 | 33 | 33 | 33 | **30** | 26 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 29 | 19 | 10 | -5 | -18 | -27 | **-10** | 54 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 29 | 32 | 35 | 37 | 39 | 40 | **39** | 39 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 29 | 31 | 33 | 33 | 33 | 34 | **30** | 27 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 29 | 34 | 43 | 43 | 42 | 42 | **34** | 27 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 29 | 32 | 36 | 37 | 37 | 37 | **33** | 29 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 29 | 52 | 71 | 69 | 67 | 65 | **46** | 29 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 29 | 49 | 69 | 68 | 66 | 64 | **47** | 30 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 29 | 44 | 45 | 46 | 46 | 47 | **42** | 37 |
| **PV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 77 | 79 | 78 | 80 | 78 | 79 | **52** | 62 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 77 | 48 | 24 | -13 | -42 | -64 | **-18** | 128 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 77 | 83 | 86 | 91 | 92 | 95 | **68** | 93 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 77 | 78 | 77 | 79 | 77 | 78 | **51** | 61 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 77 | 98 | 134 | 133 | 129 | 128 | **77** | 82 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 77 | 78 | 77 | 79 | 77 | 78 | **51** | 61 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 77 | 133 | 173 | 168 | 159 | 155 | **81** | 69 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 77 | 128 | 180 | 175 | 166 | 162 | **87** | 77 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 77 | 78 | 77 | 79 | 77 | 78 | **51** | 61 |
| **PV of debt-to-revenue ratio** | | | | | | | | |
| **Baseline** | 107 | 124 | 125 | 127 | 126 | 126 | **106** | 88 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 107 | 75 | 39 | -21 | -69 | -102 | **-36** | 181 |
| A2. New public sector loans on less favorable terms in 2014-2034 2 | 107 | 129 | 137 | 144 | 148 | 152 | **137** | 131 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 107 | 124 | 127 | 129 | 128 | 128 | **107** | 89 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 107 | 137 | 167 | 165 | 161 | 159 | **121** | 90 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 107 | 130 | 141 | 143 | 142 | 142 | **119** | 99 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 107 | 208 | 277 | 267 | 256 | 246 | **164** | 98 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 107 | 196 | 270 | 261 | 251 | 243 | **165** | 102 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 107 | 174 | 176 | 178 | 177 | 177 | **148** | 123 |

**Table 2. Mozambique: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034 (concluded)**

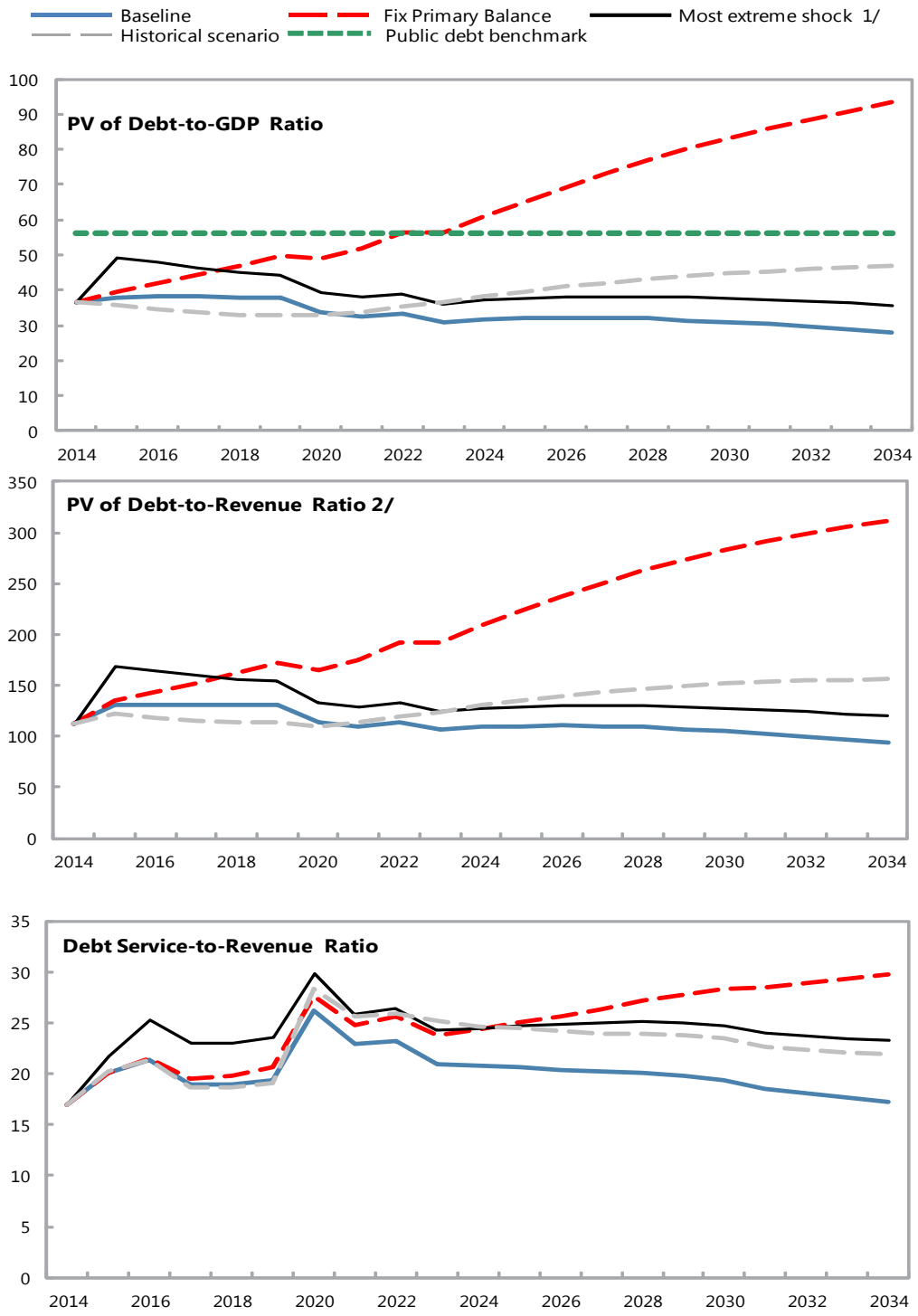
(In percent)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Debt service-to-exports ratio** | | | | | | | | |
| **Baseline** | 4 | 5 | 6 | 6 | 6 | 6 | **3** | 6 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 4 | 5 | 5 | 4 | 2 | 1 | **-2** | 5 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 4 | 5 | 6 | 6 | 6 | 6 | **4** | 8 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 4 | 5 | 6 | 6 | 6 | 6 | **3** | 6 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 4 | 6 | 8 | 9 | 9 | 8 | **6** | 9 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 4 | 5 | 6 | 6 | 6 | 6 | **3** | 6 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 4 | 5 | 8 | 9 | 9 | 9 | **7** | 8 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 4 | 6 | 9 | 10 | 9 | 9 | **7** | 9 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 4 | 5 | 6 | 6 | 6 | 6 | **3** | 6 |
| **Debt service-to-revenue ratio** | | | | | | | | |
| **Baseline** | 5 | 8 | 10 | 10 | 9 | 9 | **7** | 9 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 5 | 8 | 8 | 6 | 4 | 2 | **-3** | 7 |
| A2. New public sector loans on less favorable terms in 2014-2034 2/ | 5 | 8 | 9 | 9 | 9 | 9 | **9** | 12 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 5 | 8 | 10 | 10 | 10 | 9 | **7** | 9 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 5 | 8 | 10 | 11 | 11 | 10 | **9** | 9 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 5 | 9 | 11 | 11 | 11 | 10 | **8** | 10 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 5 | 8 | 13 | 15 | 14 | 14 | **13** | 11 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 5 | 9 | 13 | 15 | 14 | 14 | **13** | 12 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 5 | 12 | 14 | 13 | 13 | 13 | **10** | 12 |
| *Memorandum item:* | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 16 | 16 | 16 | 16 | 16 | 16 | **16** | 16 |

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the b
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after th (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

**Figure 2.Mozambique: Indicators of Public Debt Under Alternative Scenarios, 2014-2034 1/**




Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

**Table 3. Mozambique: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011-2034**

(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Estimate | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | Average 5/ | Standard Deviation 5/ | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2014-19 Average | 2024 | 2034 | 2020-34 Average |
| **Public sector debt 1/** | 39.6 | 42.7 | 53.1 | | | 56.5 | 58.3 | 58.3 | 57.7 | 56.5 | 55.5 | | 43.3 | 34.4 | |
| *of which: foreign-currency denominated* | 32.9 | 36.9 | 44.2 | | | 49.2 | 51.4 | 52.3 | 52.4 | 51.6 | 51.0 | | 41.4 | 32.8 | |
| Change in public sector debt | -6.2 | 3.1 | 10.4 | | | 3.4 | 1.8 | 0.0 | -0.7 | -1.2 | -0.9 | | 0.2 | -1.4 | |
| Identified debt-creating flows | -6.9 | 2.5 | -2.0 | | | 4.8 | 1.7 | 0.5 | -0.3 | -0.7 | -0.8 | | -0.9 | -1.8 | |
| Primary deficit | 4.1 | 3.0 | 2.0 | 3.0 | 1.1 | 7.8 | 6.0 | 5.2 | 4.3 | 3.9 | 3.5 | 5.1 | 1.6 | 0.2 | 1.0 |
| Revenue and grants | 28.6 | 28.6 | 32.9 | | | 32.4 | 29.2 | 29.2 | 29.1 | 28.9 | 28.9 | | 29.1 | 29.9 | |
| *of which: grants* | 7.8 | 5.4 | 5.4 | | | 5.2 | 4.2 | 3.5 | 3.1 | 2.8 | 2.5 | | 0.9 | 0.2 | |
| Primary (noninterest) expenditure | 32.7 | 31.6 | 34.8 | | | 40.2 | 35.2 | 34.3 | 33.4 | 32.8 | 32.4 | | 30.8 | 30.1 | |
| Automatic debt dynamics | -10.8 | -0.5 | -3.9 | | | -3.0 | -4.3 | -4.7 | -4.6 | -4.6 | -4.3 | | -2.5 | -2.0 | |
| Contribution from interest rate/growth differential | -3.2 | -2.4 | -2.8 | | | -3.8 | -4.1 | -4.2 | -4.0 | -3.9 | -3.8 | | -2.5 | -1.9 | |
| *of which: contribution from average real interest rate* | -0.1 | 0.2 | 0.1 | | | 0.3 | 0.2 | 0.2 | 0.3 | 0.3 | 0.2 | | 0.2 | 0.4 | |
| *of which: contribution from real GDP growth* | -3.1 | -2.7 | -2.8 | | | -4.1 | -4.3 | -4.4 | -4.3 | -4.3 | -4.0 | | -2.7 | -2.3 | |
| Contribution from real exchange rate depreciation | -7.6 | 1.9 | -1.2 | | | 0.8 | -0.2 | -0.4 | -0.6 | -0.7 | -0.5 | | ... | ... | |
| Other identified debt-creating flows | -0.1 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | -0.1 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | 0.6 | 0.7 | 12.4 | | | -1.4 | 0.1 | -0.4 | -0.3 | -0.5 | -0.2 | | 1.0 | 0.4 | |
| **Other Sustainability Indicators** | | | | | | | | | | | | | | | |
| **PV of public sector debt** | ... | ... | 34.5 | | | 36.4 | 38.0 | 38.2 | 38.1 | 37.8 | 37.7 | | 31.7 | 27.9 | |
| *of which: foreign-currency denominated* | ... | ... | 25.5 | | | 29.1 | 31.1 | 32.2 | 32.8 | 32.9 | 33.2 | | 29.8 | 26.3 | |
| *of which: external* | ... | ... | 25.5 | | | 29.1 | 31.1 | 32.2 | 32.8 | 32.9 | 33.2 | | 29.8 | 26.3 | |
| PV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need 2/ | 9.9 | 7.1 | 6.5 | | | 13.3 | 11.9 | 11.4 | 9.8 | 9.3 | 9.1 | | 7.7 | 5.3 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | ... | ... | 104.9 | | | 112.2 | 130.3 | 131.0 | 131.2 | 130.7 | 130.6 | | 109.1 | 93.2 | |
| PV of public sector debt-to-revenue ratio (in percent) | ... | ... | 125.6 | | | 133.4 | 152.0 | 148.8 | 147.0 | 144.7 | 143.1 | | 112.5 | 93.8 | |
| *of which: external 3/* | ... | ... | 92.9 | | | 106.7 | 124.3 | 125.3 | 126.5 | 126.0 | 125.9 | | 105.7 | 88.4 | |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 20.3 | 14.4 | 13.9 | | | 17.0 | 20.0 | 21.4 | 19.0 | 18.9 | 19.4 | | 20.7 | 17.3 | |
| Debt service-to-revenue ratio (in percent) 4/ | 28.0 | 17.7 | 16.6 | | | 20.2 | 23.4 | 24.3 | 21.2 | 21.0 | 21.3 | | 21.4 | 17.4 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | 10.3 | -0.1 | -8.5 | | | 4.4 | 4.2 | 5.1 | 5.0 | 5.1 | 4.5 | | 1.5 | 1.6 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 7.3 | 7.2 | 7.1 | 7.4 | 0.7 | 8.3 | 8.2 | 8.2 | 7.9 | 8.0 | 7.7 | 8.0 | 6.7 | 6.9 | 8.4 |
| Average nominal interest rate on forex debt (in percent) | 0.9 | 1.0 | 0.9 | 0.8 | 0.6 | 1.7 | 1.8 | 1.9 | 2.0 | 2.0 | 2.1 | 1.9 | 2.5 | 3.1 | 2.7 |
| Average real interest rate on domestic debt (in percent) | 5.5 | 7.3 | 3.7 | 3.8 | 3.3 | 3.2 | 4.3 | 4.1 | 5.4 | 6.8 | 7.8 | 5.3 | 9.2 | 9.0 | 6.6 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | -20.8 | 6.3 | -3.4 | -1.8 | 14.0 | 2.0 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 7.9 | 4.3 | 5.5 | 7.4 | 2.1 | 5.6 | 5.6 | 5.6 | 5.6 | 5.6 | 5.6 | 5.6 | 5.6 | 5.6 | 5.6 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 9.5 | 3.8 | 18.0 | 3.2 | 6.0 | 25.2 | -5.4 | 5.6 | 4.9 | 6.0 | 6.4 | 7.1 | 6.5 | 7.0 | 7.8 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | ... | ... | 37.3 | 32.4 | 30.2 | 29.0 | 26.9 | 25.6 | 30.2 | 19.4 | 10.8 | ... |

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

**Table 4. Mozambique: Sensitivity Analysis for Key Indicators of Public Debt 2014-2034**

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2024 | 2034 |
| **PV of Debt-to-GDP Ratio** | | | | | | | | |
| **Baseline** | 36 | 38 | 38 | 38 | 38 | 38 | 32 | 28 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 36 | 36 | 35 | 34 | 33 | 33 | 38 | 47 |
| A2. Primary balance is unchanged from 2014 | 36 | 39 | 42 | 44 | 47 | 50 | 61 | 93 |
| A3. Permanently lower GDP growth 1/ | 36 | 38 | 38 | 38 | 38 | 38 | 34 | 34 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-20 | 36 | 39 | 40 | 41 | 41 | 41 | 37 | 37 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-201 | 36 | 36 | 36 | 36 | 36 | 36 | 31 | 27 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 36 | 36 | 36 | 36 | 37 | 37 | 34 | 33 |
| B4. One-time 30 percent real depreciation in 2015 | 36 | 49 | 48 | 46 | 45 | 44 | 37 | 36 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 36 | 46 | 46 | 45 | 45 | 44 | 36 | 30 |
| **PV of Debt-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 112 | 130 | 131 | 131 | 131 | 131 | 109 | 93 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 112 | 122 | 118 | 116 | 114 | 113 | 130 | 157 |
| A2. Primary balance is unchanged from 2014 | 112 | 135 | 143 | 152 | 162 | 172 | 209 | 312 |
| A3. Permanently lower GDP growth 1/ | 112 | 131 | 132 | 132 | 132 | 133 | 116 | 114 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-20 | 112 | 133 | 137 | 139 | 141 | 142 | 127 | 123 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-201 | 112 | 125 | 123 | 124 | 124 | 124 | 105 | 91 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 112 | 124 | 122 | 124 | 126 | 128 | 115 | 112 |
| B4. One-time 30 percent real depreciation in 2015 | 112 | 169 | 164 | 160 | 156 | 154 | 127 | 120 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 112 | 159 | 158 | 156 | 154 | 153 | 123 | 101 |
| **Debt Service-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 17 | 20 | 21 | 19 | 19 | 19 | 21 | 17 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 17 | 20 | 21 | 19 | 19 | 19 | 25 | 22 |
| A2. Primary balance is unchanged from 2014 | 17 | 20 | 22 | 19 | 20 | 21 | 24 | 30 |
| A3. Permanently lower GDP growth 1/ | 17 | 20 | 21 | 19 | 19 | 20 | 21 | 19 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-20 | 17 | 20 | 22 | 20 | 20 | 20 | 22 | 19 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-201 | 17 | 20 | 21 | 19 | 19 | 19 | 20 | 17 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 17 | 20 | 21 | 19 | 19 | 20 | 21 | 18 |
| B4. One-time 30 percent real depreciation in 2015 | 17 | 22 | 25 | 23 | 23 | 24 | 24 | 23 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 17 | 20 | 22 | 20 | 20 | 20 | 22 | 18 |

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.